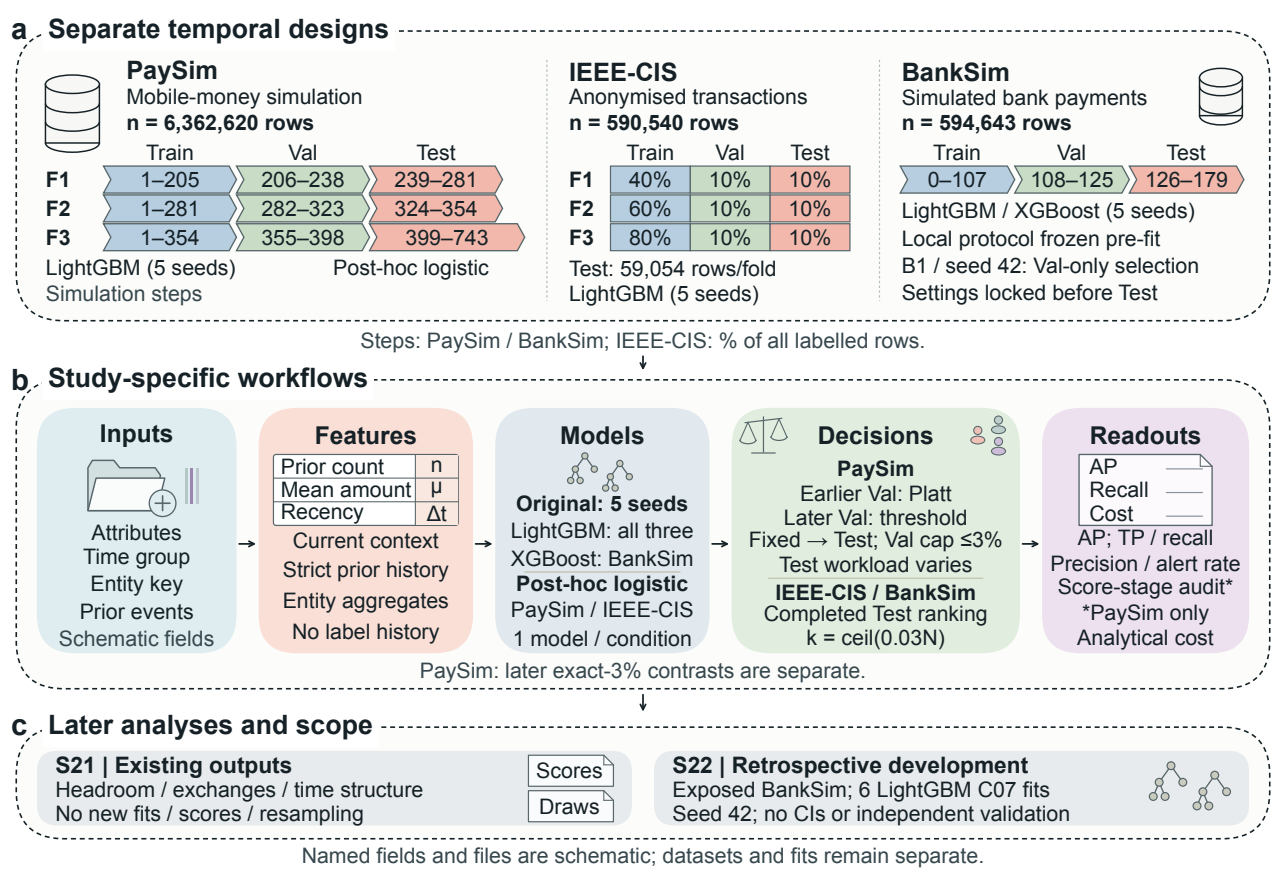

a Separate temporal designs
PaySim
Mobile-money simulation
n = 6,362,620 rows
Train Val Test
F1 1–205 206–238 239–281
F2 1–281 282–323 324–354
F3 1–354 355–398 399–743
LightGBM (5 seeds)
Post-hoc logistic
Simulation steps
IEEE-CIS
Anonymised transactions
n = 590,540 rows
Train Val Test
F1 40% 10% 10%
F2 60% 10% 10%
F3 80% 10% 10%
Test: 59,054 rows/fold
LightGBM (5 seeds)
BankSim
Simulated bank payments
n = 594,643 rows
Train Val Test
0–107 108–125 126–179
LightGBM / XGBoost (5 seeds)
Local protocol frozen pre-fit
B1 / seed 42: Val-only selection
Settings locked before Test
Steps: PaySim / BankSim; IEEE-CIS: % of all labelled rows.
b Study-specific workflows
Inputs
Attributes
Time group
Entity key
Prior events
Schematic fields
Features
Prior count n
Mean amount μ
Recency Δt
Current context
Strict prior history
Entity aggregates
No label history
Models
Original: 5 seeds
LightGBM: all three
XGBoost: BankSim
Post-hoc logistic
PaySim / IEEE-CIS
1 model / condition
Decisions
PaySim
Earlier Val: Platt
Later Val: threshold
Fixed → Test; Val cap ≤3%
Test workload varies
IEEE-CIS / BankSim
Completed Test ranking
k = ceil(0.03N)
Readouts
AP
Recall
Cost
AP; TP / recall
Precision / alert rate
Score-stage audit*
*PaySim only
Analytical cost
PaySim: later exact-3% contrasts are separate.
c Later analyses and scope
S21 | Existing outputs
Headroom / exchanges / time structure
No new fits / scores / resampling
Scores
Draws
S22 | Retrospective development
Exposed BankSim; 6 LightGBM C07 fits
Seed 42; no CIs or independent validation
Named fields and files are schematic; datasets and fits remain separate.